

Marketing Campaign Analysis

Data Cleaning, Statistics, Machine Learning & Business Insights
Report

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Executive Summary

This report analyzes a messy, multi-currency, multilingual marketing campaign dataset spanning 8 countries and 6 channels. After a fuzzy-matching cleaning pipeline resolved typos, currency formats, and geo-inconsistencies, 1033 of 1260 rows (82.0%) were retained for analysis. A machine learning model was trained to predict conversion rate from campaign attributes, and concrete budget-reallocation recommendations were derived directly from the cleaned data.

1,033	6.08%	\$13,087,827	-0.001
Rows Analyzed	Avg Conversion	Total Budget	Best Model R²

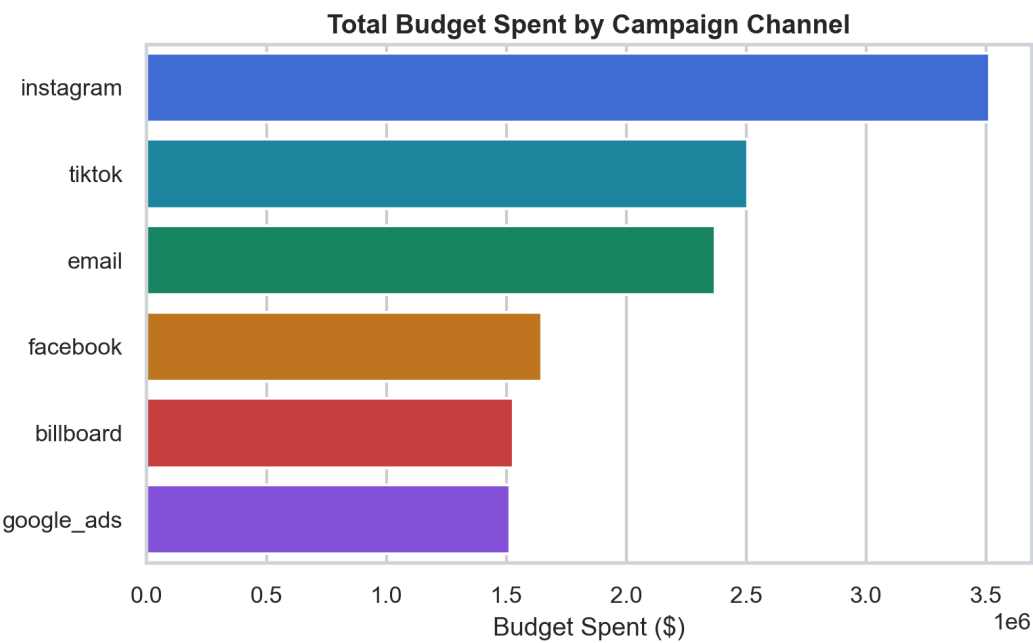
Data Quality & Cleaning Summary

The raw dataset contained 1260 rows with typos, mixed currency symbols (€/ \$), percent signs, decimal commas, duplicated/garbled category strings, inconsistent date formats, and 1-20% missing values per column. After cleaning, 1033 rows (82.0%) remained. Corrections applied: normalized text casing and accents; fuzzy-matched typo variants of country, city, product, category and channel names against a canonical vocabulary (threshold 0.72 similarity); corrected country values to match known cities; stripped currency/percent symbols and coerced numeric columns; parsed inconsistent date formats to ISO; and imputed remaining gaps via column mean (numeric) or mode (categorical). The 'impressions' column was dropped entirely — its values were too corrupted and inconsistently scaled to trust for analysis.

Rows before cleaning: 1260 → after cleaning: 1033

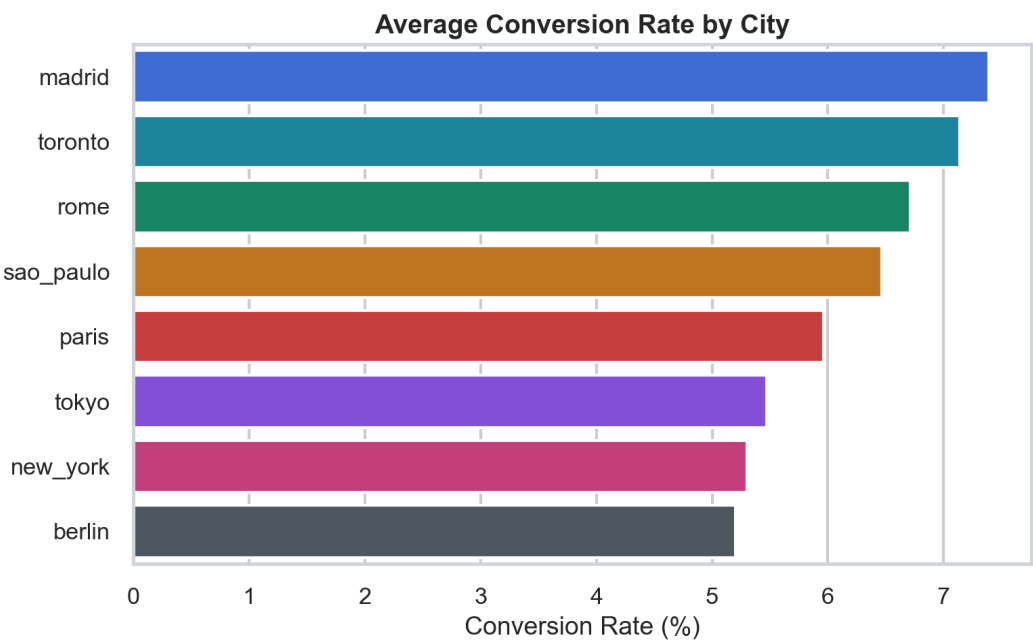
Descriptive Statistics

Budget Spent by Campaign Channel



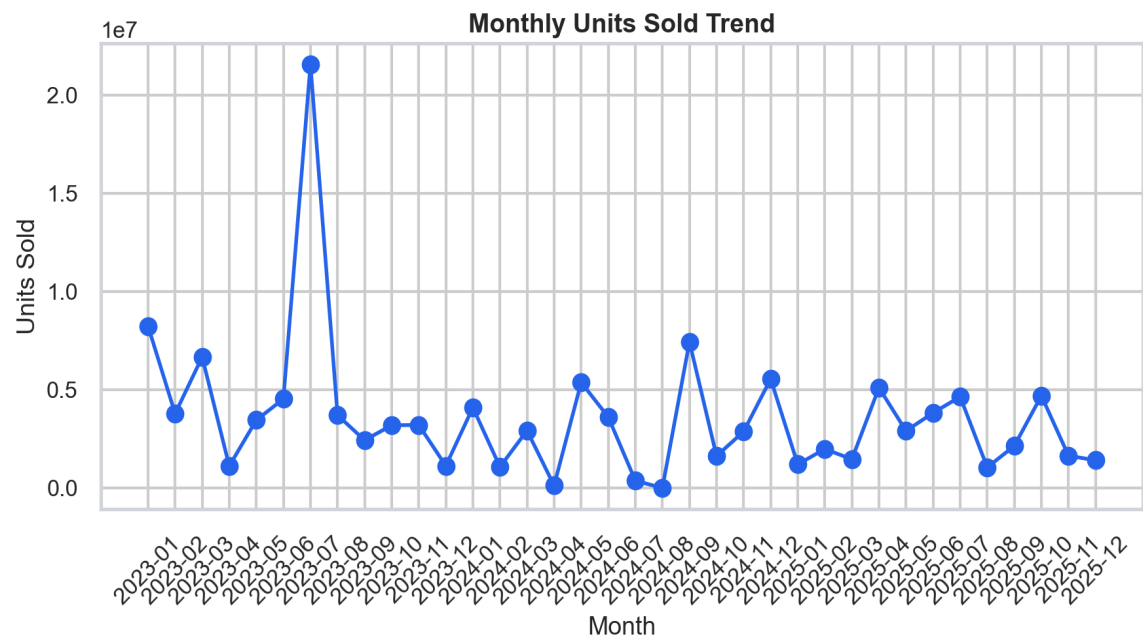
Total budget allocated across each marketing channel.

Conversion Rate by City



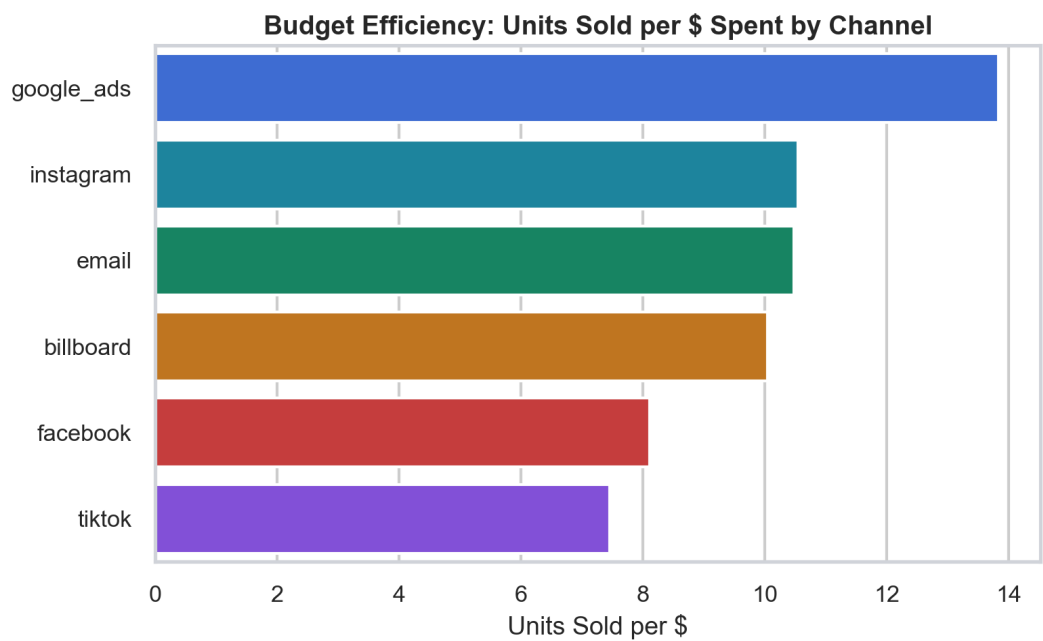
Average conversion rate per city, cleaned and geo-corrected.

Monthly Units Sold Trend



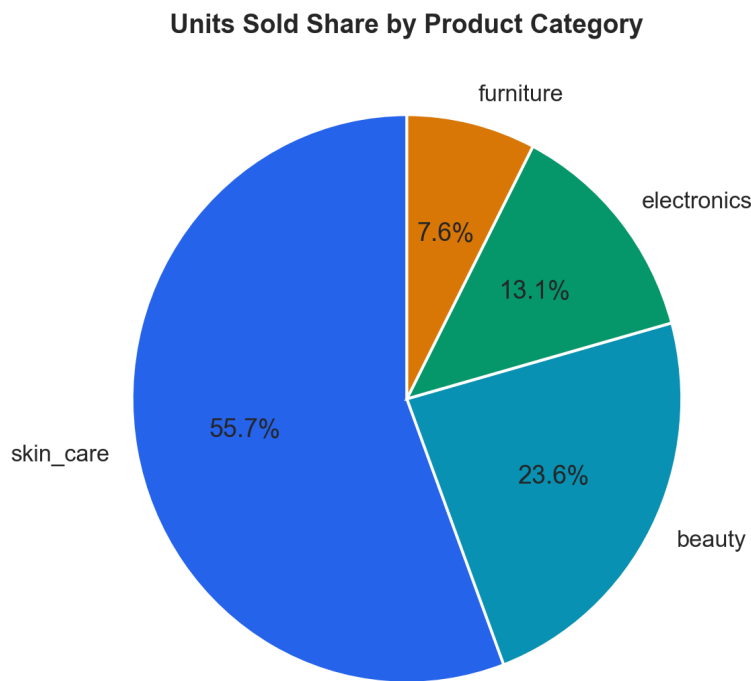
Time trend showing total units sold aggregated by month.

Budget Efficiency by Channel



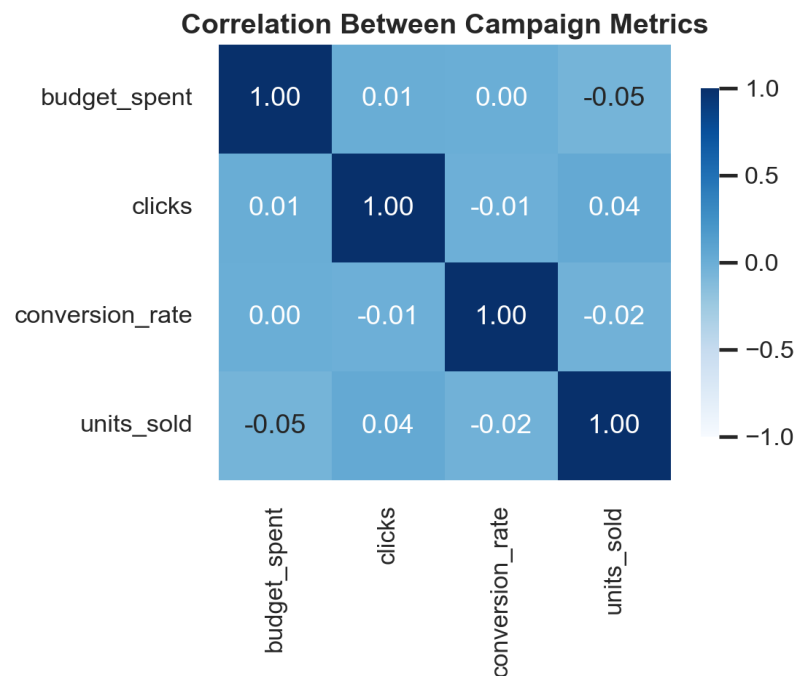
Units sold per dollar spent, ranking channels by return on budget.

Product Category Breakdown



Share of total units sold by product category.

Correlation Between Campaign Metrics



Pearson correlation across budget, clicks, conversion rate and units sold.

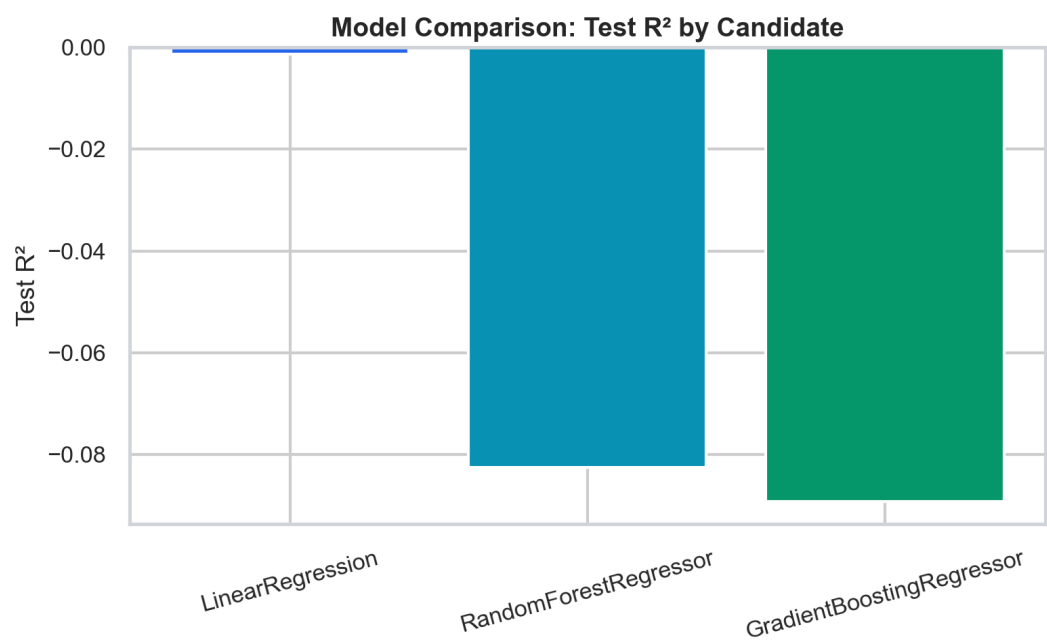
Machine Learning: Conversion Rate Prediction

A regression model was trained to predict conversion_rate from campaign attributes (country, city, product, category, channel, budget, clicks, units sold, and date-derived month/day-of-week). Three candidates were compared via 5-fold cross-validation and a held-out test set: Linear Regression, Random Forest, and Gradient Boosting.

Model	CV R ² (mean)	Test R ²	Test RMSE
LinearRegression (best)	-0.027	-0.001	7.994
RandomForestRegressor	-0.104	-0.083	8.312
GradientBoostingRegressor	-0.140	-0.089	8.337

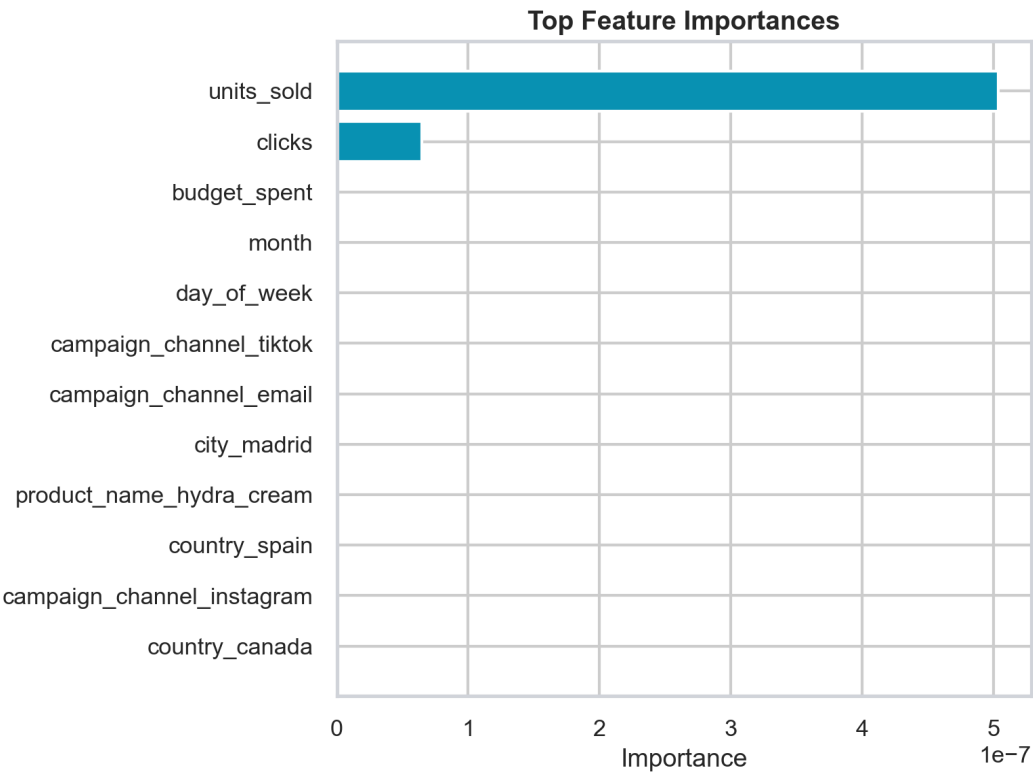
Winning model: **LinearRegression** with test R² = -0.001 and RMSE = 7.994.

Model Comparison

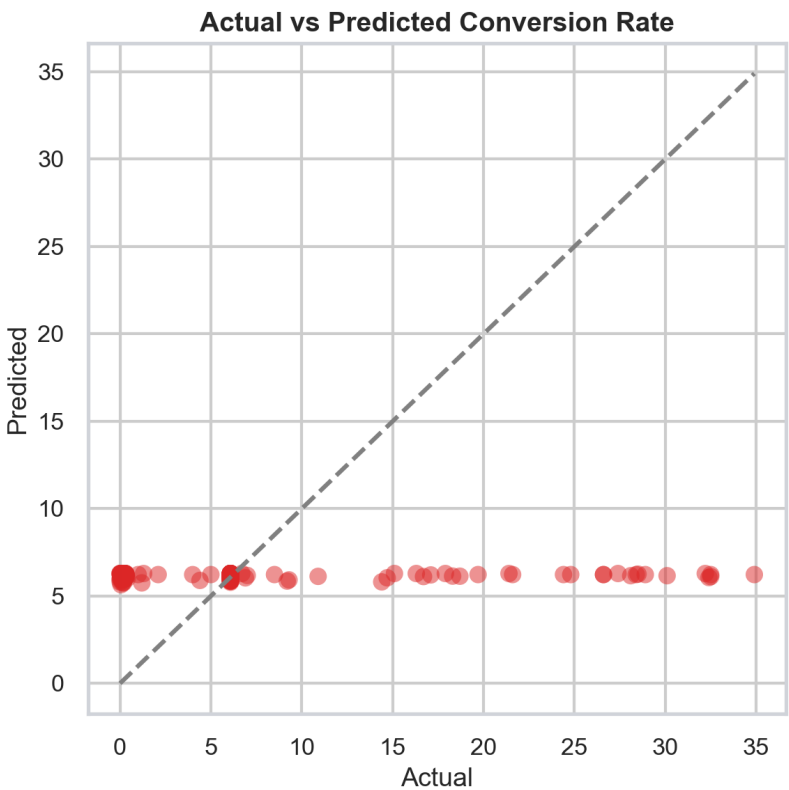


Test R² for each candidate model; higher is better.

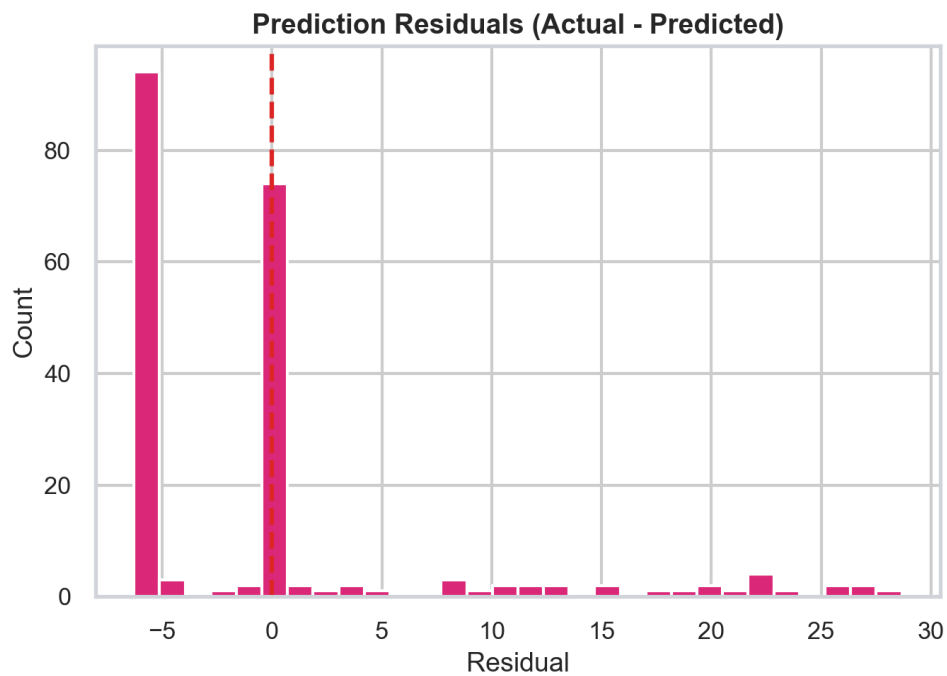
Top Feature Importances



Actual vs Predicted Conversion Rate



Prediction Residuals



Distribution of actual minus predicted conversion rate; centered on zero indicates unbiased predictions.

Business Insights & Recommendations

Best and worst channel by budget efficiency

'google_ads' delivers 13.826 units sold per \$ spent, the highest of all channels, while 'tiktok' trails at 7.445 units per \$ — a 1.9x efficiency gap.

Highest converting channel

'instagram' has the highest average conversion rate at 6.55%, versus a dataset average of 6.08%.

Top converting cities

Top 3 cities by average conversion rate: madrid (7.39%), toronto (7.14%), rome (6.71%). Lowest is berlin at 5.20%.

Budget allocation vs. sales output by country

'france' receives the largest budget share (\$2,191,132), while 'japan' actually sells the most units (22,123,939). This budget-to-output misalignment suggests reallocating spend toward 'japan' could improve overall ROI.

Best performing product category

'electronics' converts best at 6.68% on average, ahead of 'furniture' at 5.73%.

Strongest numeric relationship

'budget_spent' and 'units_sold' show the strongest correlation in the dataset at $r=-0.05$.

Recommendations

1. Reallocate incremental budget toward 'google_ads', which returns 13.826 units sold per \$ spent versus 7.445 for 'tiktok'.
2. Scale creative testing on 'instagram' campaigns, which already convert at 6.55% on average, above the 6.08% dataset mean.
3. Prioritize localized campaigns in 'madrid', the top-converting city at 7.39%, to compound existing demand.
4. Investigate why 'france' absorbs the largest budget (\$2,191,132) yet 'japan' generates more units sold (22,123,939); shifting spend could raise blended ROI.
5. Retire or redesign the lowest-efficiency channel/city combinations identified above before the next budget cycle, and re-test with a smaller pilot spend.

Appendix: Methodology

Cleaning: string normalization, fuzzy canonicalization (difflib SequenceMatcher, threshold 0.72), currency/percent stripping, numeric coercion, date parsing, city-implies-country geo correction, and mean/mode imputation. Statistics: pandas groupby aggregations, pivot tables, and Pearson correlation. Machine learning: scikit-learn ColumnTransformer with one-hot encoding, 80/20 train/test split, 5-fold cross-validation, model selection by test R^2 /RMSE. Visualization: Plotly for the interactive dashboard, matplotlib/seaborn for static report charts. Report generation: reportlab platypus.